

Foundations of Trustworthy AI

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Class 8: Robustness Overview

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What Does Quadruply Robust Mean? ^{*}

- Distribution $D \in \mathcal{D}$ over domain $\mathcal{X}$
 $\mathcal{D}$ _{class of distributions}

(Supervised learning the domain is $\mathcal{X} \times \mathcal{Y}$)

- Sample $X = (X_1, \dots, X_n)$ from the distribution

(Supervised learning samples are (X_i, Y_i))

- We want to estimate a parameter $\theta \in \Theta$
- Loss function $l: \Theta \times \mathcal{D} \rightarrow \mathbb{R}$ measures how well θ fits D
- There is an algorithm $A: \mathcal{X}^n \rightarrow \Theta$, $A(X) = \hat{\theta}$ is the model or estimate

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A robust algorithm (very roughly)
is one that has small error
when $\mathcal{D}$ and $\mathcal{D}$ are
different in some way.

Want to find a parameter with loss as small as possible

$$\theta^* = \underset{\theta \in \Theta}{\operatorname{argmin}} \ell(\theta; \mathcal{D})$$

Excess error of A
on $\mathcal{D}$ is

$$\mathbb{E}_{X \sim \mathcal{D}^n} [\ell(A(X); \mathcal{D}) - \ell(\theta^*; \mathcal{D})]$$

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Expressing Learning Problems

Domain $\mathcal{X} = \mathbb{R}^d$ Parameter space $\Theta = \mathbb{R}^d$

Mean Estimation:

Goal is to estimate $\mu = \mathbb{E}[X]$
 $X \sim D$

Loss $l(\hat{\mu}; D) = \|\hat{\mu} - \mu\|_2^2$

$$\begin{aligned}\hat{\mu} &= A(X) \\ \mu &= \mathbb{E}[X] \\ D\end{aligned}$$

Covariance Estimation:

Domain $\mathcal{X} = \mathbb{R}^d$

Goal is to estimate $\Sigma = \mathbb{E} \left[\overbrace{(X - \mu)(X - \mu)^T}^{\text{column vectors}} \right]$
 $X \sim D$

Loss $l(\hat{\Sigma}; D) = \|\hat{\Sigma} - \Sigma\|_F^2$

Frobenius

l_2^2 norm of Σ flattened to $\mathbb{R}^{d^2}$

Expressing Learning Problems

Domain $\mathcal{X} = \mathbb{R}^d$

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Mean Estimation:

Goal is to estimate $\mu = \mathbb{E}[X]$
 $X \sim \mathcal{D}$

Loss $\ell(\hat{\mu}; \mathcal{D}) = \|\hat{\mu} - \mu\|_2^2$

$$\begin{aligned}\hat{\mu} &= A(x) \\ \mu &= \mathbb{E}_{\mathcal{D}}[X]\end{aligned}$$

What Distributions $\mathbb{D}$?

$$\mathcal{D}_{\text{Gauss}} = \{ N(\mu, \Sigma) : \mu \in \mathbb{R}^d, \Sigma \in \mathbb{R}^{d \times d}, \Sigma \succeq 0 \}$$

$$p_{\mu, \Sigma}(x) = \frac{1}{c_{d, \Sigma}} \cdot e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}$$

Expressing Learning Problems

Supervised learning / Classification:

$$\ell(\theta; \mathcal{D}) = \mathbb{E}_{(x,y) \sim \mathcal{D}} [\ell(\theta; (x,y))] \quad \text{Decomposable loss}$$

$$\ell(\theta; (x,y)) = \ell(\theta; \delta_{(x,y)}) \quad \leftarrow \text{Distribution that always returns } (x,y)$$

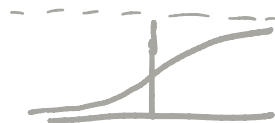
$f_{\theta}(x)$ is the predictor with parameters θ

Linear (least squares) regression:

$$(x,y) \in \mathbb{R}^d \times \mathbb{R} \quad \theta \text{ define predictor } f_{\theta}(x) = \langle \theta, x \rangle$$

$$\ell(\theta; (x,y)) = (y - f_{\theta}(x))^2$$

Expressing Learning Problems



Logistic Regression: $(x, y) \in \mathbb{R}^d \times \{0, 1\}$

parameter θ defines $f_\theta(x) = \sigma(\langle x, \theta \rangle)$

$$\sigma(t) = \frac{1}{1 + e^{-t}}$$

$$l(\theta; (x, y)) = y \cdot \log f_\theta(x) - (1 - y) \log (1 - f_\theta(x))$$

Multiclass Classification: $(x, y) \in \mathbb{R}^d \times \{1, 2, \dots, k\}$

$f_\theta(x) \mapsto$ distribution on labels ($f_\theta(x) \in \mathbb{R}^k$)

$$l(\theta, (x, y)) = -\log \left(\frac{e^{f_\theta(x)_y}}{\sum_{j=1}^k e^{f_\theta(x)_j}} \right) \quad \text{"cross-entropy loss"}$$

Expressing Learning Problems

Stochastic Optimization:

decomposable loss $\ell(\theta; \mathcal{D}) = \mathbb{E}_{x \sim \mathcal{D}} [\ell(\theta; x)]$

excess loss $\mathbb{E}_{x \sim \mathcal{D}} [\ell(\hat{\theta}(x), x) - \ell(\theta^*; x)]$

Gradient descent is usually studied at this level of abstraction

Robustness #1 ("Robust Statistics")

"Adversarial Corruption"

$$X = (X_1, \dots, X_n) \xrightarrow[\substack{\tau\text{-adversary} \\ \mathcal{C}}]{\quad} \tilde{X} = (\tilde{X}_1, \dots, \tilde{X}_n)$$

" τ -corruption adversary"
"corruption rate τ "

At most τn
points have
changed

$$\max_{\substack{\mathcal{C}: \text{corruption} \\ \text{rate } \tau}} \mathbb{E}_{X \sim D^n} [\ell(\hat{\mu}(\mathcal{C}(X)); D)]$$

Algorithm A wants to minimize

Robustness #1 ("Robust Statistics")

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" τ -corruption adversary"
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At most τn
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$$\max_{\substack{\text{c: corruption} \\ \text{rate } \tau}} \mathbb{E} [l(A(c(X)); D)]$$

Algorithm A wants to minimize

$$c(X) \neq \tilde{D}^n$$

$$D \xRightarrow{\tau\text{-adversary}} \tilde{D} = (1-\tau)D + \tau D'$$

$$\max_{\tilde{D} = (1-\tau)D + \tau D'} \left(\mathbb{E}_{X \sim \tilde{D}^n} [l(A(X); D)] \right)$$

Robustness #2 ("Distribution Shift")

$$\mathbb{E}_{x \sim D^n} [\ell(A(x); \tilde{D})]$$

Need to assume something about $D, \tilde{D}$

Adversarial Examples

$$x \mapsto x' : d(x, x') \leq \rho$$

$\uparrow$
 ρ -adversary

$$\mathbb{E}_{x \sim D^n} \left[\mathbb{E}_{(x, y) \sim D} \left[\max_{\tilde{x} : d(x, \tilde{x}) \leq \rho} \ell(A(\tilde{x}); (x, y)) \right] \right]$$

